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E-221 DC DISTRIBUTION PANELS

1. SCOPE OF WORK

The Work shall include the design, supply, manufacturing, factory testing, delivery to Site, installation, commissioning and testing of all DC Distribution Panels in accordance with Part 4, Subsection 1.2, Summary of Work.

2. CODES AND STANDARDS

The Contractor shall comply with the requirements of Part 4, Subsection 2.5 Codes and Standards, and the following specific standards.

2.1 National

- PUIL : Peraturan Umum Instalasi Listrik - Indonesia (General Regulation of Electrical Installation - Indonesia)
- SPLN : Standard Perusahaan Umum Listrik Negara

2.2 International Electrotechnical Commission

- IEC 60028 : International standard of resistance for copper
- IEC 60051 : Direct acting indicating analogue electrical measuring instruments and their accessories
- IEC 60269 : Low-voltage fuses
- IEC 60439 : Low-voltage switchgear and controlgear assemblies
- IEC 60445 : Basic and safety principles for man-machine interface, marking indication.
- IEC 60947 : Low-voltage switchgear and controlgear

2.3 American National Standards Institute

- ANSI C37.20 : Standard for Switchgear Assemblies and Metal-Enclosed Bus.

3. OPERATING AND DESIGN CONDITIONS

3.1 General

The DC distribution panels included in this Specification will supply DC power for uninterruptible power supplies (UPS), DC motors, emergency lighting, circuit

breaker control, plant control and instrumentation and data logging. Power will be supplied to the equipment from two (2) sources, battery chargers and batteries.

3.2 Equipment Ratings

220V DC Distribution Panel

3.2.1 Circuit Breaker Ratings

(1)	Number of poles	:	2
(2)	Nominal system voltage	:	220 V DC
(3)	Rated voltage	:	250 V DC
(4)	Rated normal current		
	a. Main breakers	:	as required*)
	b. Feeder breakers	:	as required*)
(5)	Rated short-circuit breaking current	:	20 kA*)
(6)	Rated duration of short-circuit	:	one (1) second

3.2.2 Bus Ratings

Continuous rating	:	1600 A*)
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3.2.3 Transfer Switch Rating

(1)	Number of poles	:	2
(2)	Nominal system voltage	:	220 V DC
(3)	Rated Voltage	:	250 V DC
(4)	Continuous rating	:	600 A*)
(5)	Momentary rating	:	20 kA*)
(6)	Rated duration of short circuit	:	one (1) second

Note: *) Item to be determined by the Contractor and subject to approve by the Owner.

4. DETAILED REQUIREMENTS

4.1 General

Distribution panel shall be manufacture or assembled Indonesia. The DC distribution panels shall be compartmented, ventilated, indoor type, free standing

metal enclosures. The metal thickness shall be according to technical specification E-280. They shall be sectionalized to contain buses and connections, wiring, circuit breakers, control circuits and devices and distribution panel as required.

All bare bus and power connections (except the ground bus) shall be covered to prevent accidental contact by personnel.

4.2 Structures

Supporting frames with sheet metal enclosures shall be designed and fabricated to form rigid self-supporting structures. Each compartment shall have a hinged gasketed door.

The back of each vertical group of compartments shall be provided with a hinged and gasketed door.

Anti-condensation space heaters with thermostatic control shall be provided in accordance with the Technical Specification E-001.

Padlocking facilities or other approved type of locking devices shall be provided to prevent unauthorized operation or access to the equipment. Locking facilities shall also be provided on cubicle access doors.

DC distribution panels shall be fitted with an approved telephone jack-in socket.

All openings on indoor distribution panels shall be protected with drip-proof louvers. The distribution panels shall have a degree of protection of at least IP 42 for indoor.

A 7 x 50 mm copper ground bus shall be installed in each structure and bonded or bolted to all parts of the structure.

4.3 Buses

Main buses, feeder buses, ground bus and connection between the several pieces of apparatus forming the equipment of a cubicle shall be of high conductivity copper in accordance with IEC 60028, and have a conductivity of not less than 99.9% of that standard.

Main buses shall be contained in a separate compartment within the panel and access shall only be possible by means of bolted-on sheet steel covers. Main buses and bus connections shall not be exposed when covers and doors are opened for access to the remainder of the panel.

All buses, connections and their supports shall be capable of carrying the short-time current associated with their short-circuit ratings for a period of one (1) second.

Bolted bus and bar joints and main and auxiliary contacts shall be silver plated.

4.4 Circuit Breakers

Incoming breakers shall be two-pole manually operated drawout air magnetic circuit breakers with a breaking current rating of *) kA at 250 volts DC on an O-CO basis, as equipped with short-time selective trip devices. These breakers shall be equipped with a long-time-delay overcurrent device on each pole adjustable from 80% to 160% of rating. Each breaker shall be equipped with a short-time-delay selective overcurrent device in each pole adjustable from 400% to 1000% of rating with time delay adjustable from 5 to 25 cycles. The breaker shall not be equipped with instantaneous overcurrent devices. The breakers shall be equipped with a normally-open bell-alarm contact to indicate an automatic trip. (*Item to be determined by the Contractor and subject to the Owner's approval.)

All outgoing breakers shall be two-pole, manually operated, drawout type, air magnetic circuit breakers with a breaking current rating of *) kA at 250 volts DC on an O-CO basis. Each breaker shall be equipped with a long-time-delay overcurrent device and an instantaneous overcurrent device on each pole. Each breaker shall be equipped with a normally-open bell-alarm contact to indicate an automatic trip. (*Item to be determined by the Contractor and subject to the Owner's approval.)

All outgoing breakers used in combination starters for motor service shall have adjustable magnetic trip only, in place of the inverse time thermal overload and instantaneous overcurrent trips.

Circuit breakers shall have a manual operating handle on the front of the compartment door. The position of the handle shall indicate whether the breaker is

open or closed. Provision shall be made for locking the breaker in the open position with two (2) or more padlocks.

4.5 Distribution Panels

The Contractor shall provide DC distribution panels and sub distribution panels as necessary to supply DC power to all items of plant which require DC power.

Distribution panels shall be provided with top and bottom removable metal gland plates which shall be drilled on site to suit the cable glands or conduit. All top entry points shall be sealed to prevent the ingress of moisture. All gland plates shall be fitted with gaskets.

Each distribution panel shall be fitted with an internal ground bus and a ground pad located in an accessible position on the outside of the panel. All metal parts of the panel except current carrying parts shall be bonded together and to the ground bus.

The distribution panel shall have removable trim, a hinged door with combination catch and lock with projecting handle.

4.6 Transfer Switches (If required)

Transfer switches for incoming feeders in the panels from the swing chargers shall be two-pole with mechanically transfer facilities to an alternate power source upon failure of its regular source and mechanically interlocked so that it shall not be possible for the load circuits to be connected to regular and alternate source simultaneously.

The transfer switches shall be fully rated to protect all types of loads from loss of continuity of power, without derating, either open or enclosed. The transfer switches shall provide complete protection with field adjustable solid state voltage sensing logic to monitor each circuit of the normal power supply.

The close differential adjustment shall be factory set to drop out when the monitored voltage drops below 70% of normal and initiate load transfer when the emergency source becomes available. Upon restoration of the normal source to a pickup level of 90%, the logic shall initiate automatic re-transfer of the load circuits to the normal power source. The transfer switches shall obtain its operating current from the source to which the load is being transferred.

The transfer switches shall have withstand, closing and interrupting rating sufficient for voltage of the system and the available short circuit at the point of the application on the drawings attached.

It shall not be possible for load circuits to be connected to normal and emergency sources simultaneously, regardless of whether the switch is electrically or manually operated.

4.7 Grounding

Each panel shall be provided with a high conductivity copper grounding bus, which shall run the full length and be bolted to the frame of the panel assembly. Grounding connections to the station grounding system and/or building steel shall be provided at each end of the bus. The ground bus dimensions shall be in no case be less than 7 x 50 mm copper or equivalent.

4.8 Control Devices and Wiring

Control relays, auxiliary contacts and small mechanism shall be enclosed, protected and accessible for maintenance. Fuses shall be located in areas where it will not be hazardous to replace them.

All internal wiring shall be only connected to one (1) side of the terminal blocks. Other side shall be used for the external connections.

Wiring and schematic diagrams covering all electrical equipment that is factory wired as an integral part of this Technical Specification shall be provided.

4.9 Indicating Instruments

All indicating instrument shall comply with IEC-60051.

Instrument scales shall be clearly divided and indelibly marked. They shall be so arranged that the normal working indication is between 50% and 75% of full scale deflection. The dials of ammeters for motor drives shall include a red line or marker to indicate the full load current of the motor. The meter dials shall have white backgrounds with black expanded scales, black figures and black pointers.

Each panel shall be equipped with the following:

- (1) One (1) DC ammeter and DC voltmeter

- (2) One (1) DC voltmeter selector switch to select bus and battery voltage.
- (3) One (1) ground detector voltmeter and push button

4.10 Nameplates

The nameplates shall be provided for all panels. The nameplate shall list all panel data required by IEC 60947.

Each panel shall be provided with a main label and danger/caution label where applicable. Main labels shall be engraved in both the English and the Indonesian languages.

Circuit identification labels shall be provided at the front and rear of all units and an identical label shall be fitted to the front of all withdrawable moving portions. The wording shall be in the English language only.

The size of danger labels shall be such that they are easily readable. The wording shall be in both the English and the Indonesian languages.

5. PERFORMANCE TESTING AND GUARANTEES

5.1 General

The Contractor may submit test certificates of type tests which have been carried out on identical equipment.

The type tests prescribed shall be carried out at the Contractor's expense in all cases where such certificates are not available or rejected.

5.2 Shop Tests

Full tests shall be made by the manufacturer on the complete equipment in accordance with the tests specified in the relevant IEC recommendations. The tests shall also include the followings:

- (1) Continuity and polarity tests on all circuits.
- (2) Functional tests on all relays.
- (3) Check of electrical and mechanical interlocks.

5.3 Site Tests

The tests shall include, but not be limited to, the followings:

- (1) Mechanical tests.
- (2) Energizing of all circuits.
- (3) Check operation of all relays, instruments, alarms and trips.
- (4) Functional testing of all equipment.

6. QUALITY ASSURANCE/QUALITY CONTROL

Quality assurance/quality control shall conform to the requirements included in Part 4, Subsection 2.7. These shall include, but not be limited to, the followings:

- (1) Material Certification.
- (2) Continuity and Wiring Tests and short Circuit Tests.
- (3) Insulation and High Voltage Tests.
- (4) Mechanical Tests and functional Tests.
- (5) Packaging and Shipping.
- (6) Welding, Cleaning and Painting.

The quality assurance/quality control documentation as specified shall include, but not be limited to, the followings:

- (1) Material Certificates.
- (2) Shop Test Reports.

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